

Authors

Dr. Max Feinberg

Paris
France

Prof. Serge Rudaz

Section des sciences pharmaceutiques
Université de Genève
Rue Michel Servet 1
1211 Genève
Switzerland

Cover Image: © Max Feinberg

■ All books published by **WILEY-VCH** are carefully produced. Nevertheless, authors, editors, and publisher do not warrant the information contained in these books, including this book, to be free of errors. Readers are advised to keep in mind that statements, data, illustrations, procedural details or other items may inadvertently be inaccurate.

Library of Congress Card No.: applied for

British Library Cataloguing-in-Publication Data

A catalogue record for this book is available from the British Library.

Bibliographic information published by the Deutsche Nationalbibliothek

The Deutsche Nationalbibliothek lists this publication in the Deutsche Nationalbibliografie; detailed bibliographic data are available on the Internet at <<http://dnb.d-nb.de>>.

© 2024 WILEY-VCH GmbH, Boschstraße 12,
69469 Weinheim, Germany

All rights reserved (including those of translation into other languages). No part of this book may be reproduced in any form – by photoprinting, microfilm, or any other means – nor transmitted or translated into a machine language without written permission from the publishers. Registered names, trademarks, etc. used in this book, even when not specifically marked as such, are not to be considered unprotected by law.

Print ISBN: 978-3-527-35332-3

ePDF ISBN: 978-3-527-84525-5

ePub ISBN: 978-3-527-84526-2

oBook ISBN: 978-3-527-84527-9

Typesetting Straive, Chennai, India

Contents

List of Figures *xi*

List of Resources *xv*

Preface *xvii*

Glossary of Symbols *xxi*

Acknowledgments *xxiii*

1	Quantification	1
1.1	Define the Measurand (Analyte)	1
1.1.1	Quantification and Calibration	2
1.1.2	Authentic <i>versus</i> Surrogate	3
1.1.3	Signal Pretreatment and Normalization	6
1.2	Calibration Modes	9
1.3	External Calibration (EC)	10
1.3.1	Authentic Analyte in Authentic Matrix: MMEC	10
1.3.2	Authentic Analyte in Surrogate Matrix	12
1.3.3	Surrogate Calibrant in Authentic Matrix	13
1.3.4	Surrogate Calibrant in Surrogate Matrix	14
1.4	In-sample Calibration (ISC)	15
1.4.1	Authentic Analyte: Standard Addition Method	15
1.5	Some New Quantification Techniques	17
1.5.1	Isotopic Pattern Deconvolution (IPD)	18
1.5.2	Direct Internal Calibration with Labeled Calibrant (IC-SIL)	20
	References	23
2	Calibration	25
2.1	Direct and Inverse Calibration	25
2.2	Least-squares Regression Method	28
2.2.1	Straight-line Computation	28
2.2.2	Assumptions and Complements	32
2.3	Software Implementation	34
2.3.1	Ordinary Least-squares (OLS) Regression	34
2.3.2	Weighted Least-squares (WLS) Regression	37
2.4	Calibration: Special Topics	41
2.4.1	Nonlinear Calibration Curve	41

2.4.2	Misuses of Regression for Calibration	45
2.4.2.1	Coefficients of Correlation and Determination	45
2.4.2.2	Definitions of Linearity	47
2.4.3	Statistical Aspects of Standard Addition Method (SAM)	48
2.5	Metrological Approach to Calibration	51
2.5.1	Errors in Inverse-predicted Values	53
2.5.2	Calibration as a Source of Uncertainty	55
	References	57
3	Precision	59
3.1	Outputs of Interlaboratory Studies	59
3.1.1	Diverse Precision Parameters	59
3.1.2	Role of Series for Data Collection	60
3.2	Analysis of Variance (ANOVA)	64
3.2.1	Computation of Precision Parameters	64
3.2.2	Additional Parameters	69
3.2.2.1	Relative Standard Deviation of Parameters	69
3.2.2.2	Variance of the Grand Mean	70
3.3	Balanced and Unbalanced Experimental Design	71
3.4	Software Implementation	72
3.4.1	ANOVA Classic Algorithm	72
3.4.2	Detect Outliers and Stragglers	75
3.4.2.1	Other Algorithms	78
	References	79
4	Trueness	81
4.1	Trueness and True Value	81
4.1.1	Bias and Recovery Yield	82
4.1.2	Evolution of the Concept of True Value	83
4.1.3	Specificity and Sources of Bias	84
4.2	Assessment of Trueness	86
4.2.1	Primary Operating Procedures	86
4.2.2	Reference Materials	87
4.2.2.1	Certified Reference Materials (CRM)	87
4.2.2.2	External Reference Materials (ERM)	87
4.2.2.3	Internal Reference Materials (IRM)	87
4.2.2.4	Verification Standard Solutions	87
4.2.2.5	Standard Addition Method (SAM) and Surrogate Samples	88
4.3	Proficiency Testing	89
4.3.1	Interlaboratory Comparison or Proficiency Testing Scheme (PTS)	89
4.3.2	Organization of Proficiency Testing Schemes	90
4.3.3	Reference Value of the Test Material	91
4.3.4	Performance Scores	93
4.3.5	Algorithm A	94
4.3.6	Check Material Homogeneity or Stability	97

4.4	Control Charts	99
4.4.1	First Phase Assessment of the Reference Value	100
4.4.2	Second Phase Routine Use	101
	References	102
5	Method Validation	105
5.1	Review of Validation Procedures	105
5.1.1	Inconsistencies of Validation Vocabulary	107
5.1.2	Validation Plans	109
5.2	Method Accuracy Profile (MAP)	113
5.2.1	Principles	113
5.2.2	Method Accuracy Profile by Example	116
5.3	Statistical Dispersion Intervals	122
5.3.1	β -Expectation Tolerance Interval (β -ETI)	125
5.3.2	β - γ Content Tolerance Interval (β - γ -CTI)	128
5.4	Accuracy Profile: Special Topics	131
5.4.1	Choose the Best Calibration Model	132
5.4.2	Apply Consistent Experimental Design	133
5.4.3	Check the Number of Efficient Measurements	135
5.4.4	Select Probability Values	140
5.4.5	Select the Type of Tolerance Interval	143
5.4.6	Proportion of Nonacceptable Measures	144
	References	146
6	Measurement Uncertainty (MU)	149
6.1	Principle of Measurement Uncertainty	149
6.2	General Procedure to Estimating MU	150
6.3	Traceability at the International System of Units	152
6.4	Stage 1. Specify the Measurand	154
6.5	Stage 2. Identify Uncertainty Components	157
6.6	Stage 3. Quantify Uncertainty Sources	158
6.6.1	Type A Approach	159
6.6.1.1	Accuracy Profile	159
6.6.1.2	Interlaboratory Study	159
6.6.1.3	Control Chart	160
6.6.1.4	Proficiency Testing	160
6.6.2	Type B Approach	160
6.7	Stage 4. Calculate Combined Uncertainty	161
6.7.1	Law of Propagation of Uncertainty	161
6.7.1.1	The Model Only Contains Additions and Subtractions	162
6.7.1.2	The Model Only Contains Products and Quotients	163
6.7.1.3	The Model is a Complex Combination of Input Quantities	163
6.7.2	Kragten Iterative Algorithm	165
6.8	Calculate Expanded Uncertainty	169
6.9	Round the Result	170

6.10	Accuracy, Total Error, and Uncertainty	171
6.11	Insights on Probability	174
	References	177
7	Measurement Uncertainty in Analytical Sciences	179
7.1	Published Procedures: An Evaluation	179
7.2	Use Method Accuracy Profile Data	181
7.2.1	Stage 1. Generic Measurement Model	181
7.2.2	Stage 2. Generic Cause-to-Effect Diagram	182
7.2.3	Main Sources of Uncertainty in the Laboratory	185
7.2.3.1	Manpower	185
7.2.3.2	Material and Handling of Items	185
7.2.3.3	Method	185
7.2.3.4	Machine/Equipment	186
7.2.3.5	Environment	186
7.2.3.6	Measurement and Other Sources	186
7.2.4	Stages 3 and 4. Calculation of Combined Uncertainty	187
7.3	Use Control Charts Data	191
7.3.1	Principles of the Shewhart Control Chart	191
7.3.2	Statistical Dispersion Intervals and Control Charts	195
7.3.3	Estimation of the Reference Value Uncertainty	198
7.4	Use Interlaboratory Comparison Data	200
7.4.1	Proficiency Testing Scheme (PTS)	200
7.4.2	Interlaboratory Studies	201
7.5	Uncertainty Functions	203
7.5.1	Horwitz's Model	203
7.5.2	Fitting the Uncertainty Function	208
7.5.2.1	How to Interpret a Power Function?	211
7.6	Concept of Coverage Interval	211
7.6.1	Origin of Coverage Interval	211
7.6.2	Coverage Interval of Given Concentration	215
7.6.3	Coverage Interval of Given Relative Uncertainty	215
7.6.4	Obtain the Limits of the Coverage Interval	216
	References	217
8	Measurement Uncertainty and Decision	221
8.1	Framework for Decision-Making	221
8.1.1	Decision <i>versus</i> Uncertainty	221
8.1.2	Specification Limits and Reference Values	223
8.1.3	Role of the Analytical Report	226
8.2	Sample Conformity Assessment	227
8.2.1	Define the Decision Rule	227
8.2.2	Guard Band Concept	229